Speed & Velocity

Exam code: 0625 & 0972

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Speed

- The **speed** of an object is defined as

Distance travelled per unit time

- Speed is a **scalar** quantity
  - This is because it only contains a **magnitude** (without a direction)
- For objects that are moving at a **constant speed**, the equation for calculating speed is:

$$v = \frac{s}{t}$$

- Where:
  - v = speed, measured in metres per second (m/s)
  - s = distance travelled, measured in metres (m)
  - t = time, measured in seconds (s)

Help me understand

Stuck on something?

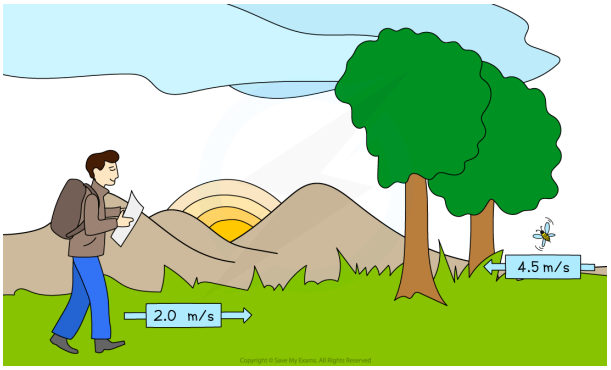
Ask questions and get help with exactly what you're revising.

Got it

Average speed

- The **speed** of an object can **vary** throughout its journey
- Therefore, it is often more useful to know an object's **average speed**

- Average running speed = 3 m/s
- Average cycling speed = 6 m/s

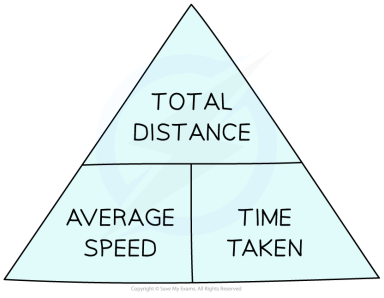


A hiker might have an average speed of 2.0 m/s, whereas a particularly excited bumble bee can have average speeds of up to 4.5 m/s

- A car travels at different instantaneous speeds throughout its journey
  - The car speeds up and slows down
  - The car may stop at junctions and traffic lights
- The average speed of the journey of the car can be found by dividing the total distance travelled by the total time taken
- The equation for calculating the **average speed** of an object is:

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

- **Average speed** considers the **total distance travelled** and the **total time taken**



Formula triangle for average speed, distance moved and time taken

How to use formula triangles

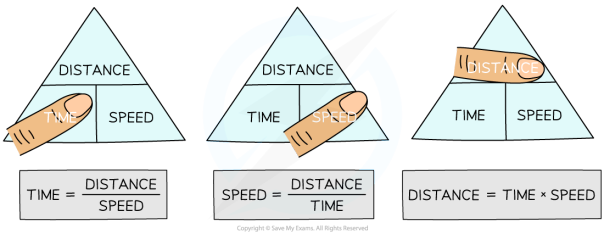
- Formula triangles are really useful for knowing how to rearrange physics equations
- To use them:

1. Cover up the quantity to be calculated, this is known as the 'subject' of the equation
2. Look at the position of the other two quantities

divided - make sure to keep the order of which is on the top and bottom of the fraction!

- In the example below, to calculate average speed, cover-up the variable speed so that only distance and time are left
  - The equation is revealed as:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$



To use a formula triangle, simply cover up the quantity you wish calculate and the structure of the equation is revealed

Worked Example

Planes fly at typical average speeds of around 250 m/s.

Calculate the distance travelled by a plane moving at this average speed for 2 hours.

Answer:

Step 1: List the known quantities

- Average speed = 250 m/s
- Time taken = 2 hours

Step 2: Write the relevant equation

$$\text{average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

Step 3: Rearrange to make distance moved the subject

$$\text{distance travelled} = \text{average speed} \times \text{time taken}$$

Step 4: Convert any units

- The time given in the question is not in standard units
- Convert 2 hours into seconds:

$$2 \text{ hours} = 2 \times 60 \times 60$$

$$2 \text{ hours} = 7200 \text{ s}$$

Step 5: Substitute the values for average speed and time taken

$$\text{distance travelled} = 250 \times 7200$$

Examiner Tips and Tricks

Rearranging equations is an important skill in Physics. You can use the equation triangles to help you practice, but it is better not to rely on them because they do not work for all equations you may need to rearrange in the exam.

Velocity

- **Velocity** is a vector quantity with magnitude and direction
- Velocity is defined as:

Speed in a given direction

- The direction of a velocity can be given in **words**
  - For example, 20 m/s east
- Or the direction of velocity can be given using a **positive** or **negative value**
  - For example, –20 m/s
- A positive direction is typically in the direction of the initial motion, to the right, or upward
- A negative velocity is typically in the opposite direction to the initial velocity, to the left, or downward

Comparing speed and velocity

The cars in the diagram above have the same speed (a scalar quantity) but different velocities (a vector quantity). Fear not, they are in different lanes!

Examiner Tips and Tricks

The positive and negative values of velocity can be assigned to any direction as long as the negative velocity is in the opposite direction to the positive value. You can decide which direction you assign to be positive as long as you are consistent throughout a question.

- The equation for velocity is very similar to the equation for speed:

- Where:
  - v = velocity in metres per second (m/s)
  - s = displacement, measured in metres (m)
  - t = time, measured in seconds (s)
- Velocity is a vector quantity, so it uses displacement, s, which is another vector quantity

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